

# EC325: Econometrics

Colby College | Spring 2026

Prof. Ray Caraher

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  - Office: 365 Diamond Building
  - Office Hours:
    - Tue: 2:30pm - 4:00pm
    - Thu: 9:30am - 10:30am; 2:30pm - 4:00pm
    - Fri: 10:00am - 12:00pm
    - Or by [appointment](#)
  - Class Meeting Days: Tue/Thu
  - Class Times: 11:00am - 12:15pm
  - Classroom: 241 Diamond Building
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## Course Description

This course explores the measurement and evaluation of economic models, bridging economic theory with empirical analysis. The emphasis throughout is on building intuitive understanding and applying econometric techniques to real-world problems encountered in public policy. With a focus on *causality* as our guiding theme, we begin by establishing a framework for causal inference based on the gold standard of randomized trials—a foundation that will inform our approach throughout the course. We then introduce the workhorse of econometric analysis: the linear regression model. You'll learn about its statistical properties and limitations, and how to interpret and communicate econometric results effectively. Building on this foundation, we explore how to apply these tools to real-life scenarios, covering categorical variables, non-linear relationships, binary outcomes, and other specialized models. The course concludes with advanced econometric techniques for analyzing two particularly powerful types of data: panel data and time series data. Econometrics demands a diverse skill set encompassing mathematics, statistics, economic theory, programming, and written and verbal communication. Perhaps most importantly, it requires developing strong intuition. This course will equip you with the analytical tools and empirical methods essential for modern economic research and policy analysis.

## Learning Objectives

### 1. Understand Core Econometric Techniques

- Develop a solid understanding of major econometric concepts and their applications in analyzing social science data

### 2. Distinguish Correlation from Causation

- Explore techniques to identify and evaluate causal relationships in data, differentiating them from mere correlations, with applications in policy evaluation

### 3. Gain Proficiency in Statistical Software

- Build hands-on experience with modern tools for data analysis, visualization, and maintaining reproducible research workflows.

### 4. Apply Econometric Techniques to Real Questions

- Utilize econometric methods to analyze real-world datasets, including in your own project, drawing meaningful conclusions and policy implications

### 5. Communicate Analytical Results Clearly

- Learn to effectively present, interpret, and document econometric findings

## Contacting Me

Please reach out if you have questions or need assistance! The best way to contact me is via email. I will do my best to respond within 24 hours on weekdays.

Especially for an econometrics course, it can be difficult to answer questions solely over email. I strongly recommend you come to my office hours or schedule an appointment if you have a question that requires more than a few sentences to answer. If the office hours listed above do not work for you, please schedule an appointment with the link above.

## Course Materials

This class will pull from three primary texts. I recommend you obtain all three, but *none are required*.

### 1. [Introductory Econometrics: A Modern Approach](#) by Jeffrey M. Wooldridge (2019)

- The 7th edition is the most recent, but any edition 5 or above will work. We will primarily use this text for part of the course more reliant on statistical theory.

### 2. [Real Econometrics](#) by Michael Bailey (2020)

- This is a good, more “applied” version of the Wooldridge textbook.

### 3. [Mastering 'Metrics](#) by Joshua D. Angrist and Jörn-Steffen Pischke (2015)

- This text is an excellent companion to the other texts, but is not like a traditional textbook.

You will also need reliable access to a computer (such as your personal laptop) to use the following software:

- [R](#), an open-source software for statistical analyses and graphics
- [R Studio](#), an integrated development environment (IDE), which provides an interface for working with R
- [Git](#), version control software used to download and update course materials

The first week of class we will go over how to install these software.

The most important set of materials you need are the course lecture notes, which you are required to read each week. These lecture notes include built-in practice questions, and you are expected to come to class having reviewed these notes.

All course materials excluding assignments you turn in—including lecture notes, datasets, and in-class labs—are hosted on the course GitHub repository at [github.com/rpcaraher/ec325](https://github.com/rpcaraher/ec325). You will clone this repository to your computer during the first week of class. When I update the materials throughout the semester, you will “pull” the latest version to keep your local copy up to date. Instructions for setting up and using the repository are included in the **README** file. Your own work can (optionally) be saved in the **student-work/** folder, which is not tracked by Git and will not be overwritten when you pull updates.

You can also access the lecture notes here: <https://www.raymondcaraher.com/ec325/>.

## Prerequisites

I expect you to have completed the following courses or their equivalents:

- EC 223: Microeconomic Theory (recommended)
- EC 225/293: Research Methods and Statistics (required)

If you have not taken these courses please speak with me the first week of class to discuss whether you are prepared for this course. It is especially important that you are comfortable with the content of EC 293.

## Assessments

### Concept Checks

This is a fast-paced, difficult course which covers a huge amount of material in a short period of time. As a result, it is very important to stay on top of the material and practice econometrics as often as possible. To assist you in doing so, each week, you will be required to complete a short set of questions, called Concept Checks, which tests your understanding of the concepts covered in the readings and lectures, and your ability to implement them with code. These will be due each Friday at 11:59pm at the end of each week (with some exceptions to be noted), although you are welcome to complete them earlier in the week if that is preferred. These will be graded on a 3 point scale (0 = no attempt, 1 = needs improvement, 2 = satisfactory, 3 = excellent). There will never be extensions granted for missed Concept Check. However, your lowest 2 grades will be dropped at the end of the semester. Be sure to use these wisely!

These concept checks must be completed individually, and there can be no collaboration with your classmates or AI tools. However, you are allowed to ask me questions via email or come to office hours. There is no time limit to complete each concept check, but they will be designed to take about 30-60 minutes per week.

## **Problem Sets**

In addition to the weekly concept checks, there will be 4 problem sets during the semester. These problem sets are designed to help you understand how econometricians use the methods we learn in class to answer interesting social questions by having you replicate results from major papers published in economics. These will be due on Moodle by 11:59pm on the specified date. They are worth a considerable portion of your final grade, so you should consider each problem set as a mini-project and begin working on it well before its due date.

## **Exams**

There will be two exams in this course: a midterm and a final. Both exams will be in-person, and you are permitted to bring one double-sided note sheet as a reference. These exams will primarily be conceptual, and you will not use software, though you may be required to analyze output from a statistical software package. The midterm exam will (tentatively) take place as noted on the syllabus during the standard class meeting time. The final exam will take place according to the slot scheduled by the registrar. More information about these exams will be provided in class.

## **Engaged Learning**

Active and engaged learning participation in class are crucial for your success in this course, and thus count as a percentage of your overall grade. You can demonstrate engaged learning through a multitude of ways, including asking and answering questions in class, coming to office hours, coordinating group members, and helping your classmates with the course material.

## **Final Project**

The capstone of this course will be a research project to be completed in groups of 3-4 students. This project will require you to apply the econometric tools introduced in class to analyze a social science phenomenon, topic, or policy of your choice. Each group will be required to submit a written report and give a presentation to the class during the final week of the semester. This project will be a considerable portion of your overall grade and will require significant time and effort to complete successfully. To help facilitate your work on the final project, we will have several check-in deadlines throughout the semester.

This includes:

- A 1 page project proposal due in week 5
- A data summary due in week 8
- A preliminary findings report due in week 10
- A final presentation during the last week of class

- A final written report due during the last day of finals
- A peer review that evaluates your personal contribution as well as your group mates

Each group will only submit one version of each deliverable. Early in the semester, I will distribute a survey asking you to rank economics subfields that interest you for your project. Based on your responses, I will form groups of students with similar interests. Each group will then select a specific topic within their assigned subfield.

At the end of the semester, you will also be required to complete a peer review survey, which asks about each group member individually about their contributions to the project, as well as about the other group members' contributions to the final project. The responses to the peer review survey will be considered in the determination of the Engaged Learning grade.

## **Grading**

The standard grading scale will be used. I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 30% of your grade will be determined by the problem sets
- 10% will be determined by the midterm exam
- 15% will be determined by the final exam
- 30% will be determined by the final project
- 10% will be determined by your concept checks
- 5% will be determined by your engaged learning

## **Course Policies**

### **Attendance Policy**

Unless otherwise noted, all classes will be in-person and attendance is required. If you have an extracurricular event which requires your missing a class, please notify me at least one week ahead of time. Additionally, do not come to class if you are feeling unwell. You are not required to produce a formal doctors note, but please let me know via email to check-in and help prevent you from falling behind. Any unexcused absences (i.e., absences not due to extracurricular events and illness) will count against your engaged learning grade.

### **Assignment Submission**

Please submit all your assignments as either Microsoft Word files (.doc/.docx) or PDF files. All files must be submitted via Moodle (do not email them to me), generally at 11:59pm on the date they are due. You must confirm that your files can be opened and are not corrupt before you turn them in.

## Late Assignments

All assignments must be completed on time. Only under the most dire circumstances will an extension be provided for a problem set. Under no circumstances will I accept a late Concept Check. If there are extenuating circumstances that prevent you from completing a component of the final project on time, please reach out to me as soon as possible to discuss your options.

## AI Policy

AI tools such as ChatGPT, Gemini, Claude, and others may be useful in helping you learn econometrics, especially with questions related to coding. However, fully understanding the material and the process of coding is essential for success in this course and your long-term career as an econometrician. Over-reliance on AI tools may hinder your ability to get the most out of your learning experience and develop critical thinking skills, which are as essential for the art of econometrics as any other field. This is something that is seen too often in this course; for students who lean on AI too much, it quickly becomes a crutch, and even the most basic tasks cannot be completed without the use of these tools. I do not want to prohibit the use of potentially useful learning tools, but I also want to mitigate the risk of over-reliance on them. Therefore, the course policy regarding the use of AI tools is as follows:

1. You are allowed to use AI tools on your **problem sets and in your final project to troubleshoot specific, code-related questions only**. No AI tools are allowed on the Concept Checks. If you find yourself tempted rely on AI for the Concept Checks, it is a sign that there is a serious gap in your understanding of the course material and you should get in touch with me about the issues you are having. All functions you will need to complete the assignments will be covered in the readings, assignments, or lectures. In addition, there is a reference section at the end of the lecture notes which includes all functions necessary to complete this course, and likely your final project, with examples relevant to the course materials. Under no circumstances can you directly ask the AI tool to fully complete a question on any assignment. Examples of acceptable uses of AI will be shown during class.
2. Any use of an AI tool must be attributed to the tool in your submission. Since AI-generated materials cannot be retrieved by graders and are not attributable to a specific individual, students should cite the creator of the AI tool (e.g., cite OpenAI when directly quoting ChatGPT).
3. You must **understand** the AI-assisted code you submit, and you must be prepared to explain it in your own words if prompted.

If you are unsure if a use case of an AI tool is allowed, you should ask the instructor for clarification. **Any use of AI outside of these parameters or without approval from the instructor will be considered a violation of Colby's Academic Honesty policies.** If you choose to use AI, I strongly encourage you to use Google Gemini with your Colby College account. Colby has worked with Google to help keep your data secure when interacting with this tool. Review the resources at the [Davis Institute for AI](#) for more information.

## Academic Integrity and Honesty

Honesty, integrity, and personal responsibility are cornerstones of a Colby education and provide the foundation for scholarly inquiry, intellectual discourse, and an open and welcoming campus community. These values are articulated in the Colby Affirmation and are central to this course. You are expected to demonstrate academic honesty in all aspects of this course. If you understand our course expectations, give credit to those whose work you rely on, and submit your best work, you are highly unlikely to commit an act of academic dishonesty. Academic dishonesty includes, but is not limited to: violating clearly stated rules for taking an exam or completing homework; plagiarism (including material from sources without a citation and quotation marks around any borrowed words); claiming another's work or a modification of another's work as one's own; buying or attempting to buy papers or projects for a course; fabricating information or citations; knowingly assisting others in acts of academic dishonesty; misrepresentations to faculty within the context of a course; and submitting the same work, including an essay that you wrote, in more than one course without the permission of the instructors. Academic dishonesty is a serious offense against the college. Sanctions for academic dishonesty are assigned by an academic review board and may include: failure on the assignment, failure in the course, or suspension or expulsion from the College. For more on recognizing and avoiding plagiarism, see: [libguides.colby.edu/avoidingplagiarism](https://libguides.colby.edu/avoidingplagiarism). For resources and information on academic integrity, see: <https://www.colby.edu/academics/academic-integrity/>.

## Accommodations for Disabilities

I am committed to creating a course that is inclusive in its design. If you encounter barriers, please let me know immediately so we can determine if there is a design adjustment that can be made. I am happy to consider creative solutions as long as they do not compromise the intent of the assessment or learning activity. If you are a student with a disability, or think you may have a disability, you are also welcome to initiate this conversation with the Dean of Students Office. The Dean of Students Office works with students with disabilities and faculty members to identify reasonable accommodations. Please visit their website for contact and other information: <https://www.colby.edu/studentadvising/student-access-and-disability-services/>. If you have already been approved for academic accommodations, please connect within the two weeks of the start of the semester so the office can develop an implementation plan.

## Mental and Emotional Health

I am invested in the mental and emotional health of my students. Even as I establish and maintain the academic standards of my course, I value each of you as individuals with complex lives, identities, and challenges. Throughout the semester, the responsibilities of your Colby education may interact with situational as well as ongoing mental and emotional challenges in foreseeable and unforeseeable ways. If you are in need of reasonable flexibility due to an emotional situation or an ongoing mental health issue, please communicate as openly as possible with your Class Dean, and/or members of the office of Access and Disability Services, preferably in advance of the need, so that we can discuss how your circumstances interface with course requirements. Together, we will consider what is needed and what is possible. If we can discuss the situation, we can manage the situation together. Please do not allow academic responsibilities to prevent you from getting help you need. Our Colby

Counseling Services staff (207-859-4490) and the staff in the Dean of Studies office (207-859-4560) are available to connect with you. The safety of my students and every member of this community is paramount. If you or someone you know is struggling with thoughts of suicide or may be a danger to themselves or others, please call the on-call counselor immediately (207-859-4490, press '0').

### **Respect for Diversity**

It is my intent that students from diverse backgrounds and perspectives be well-served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. I expect you to feel challenged and sometimes outside of your comfort zone in this course, but it is my intent to present materials and activities that are inclusive and respectful of all persons, no matter their gender, sexual orientation, disability, age, socioeconomic status, ethnicity, race, culture, perspective, and other background characteristics. Class rosters are provided to each instructor with the student's legal name. I will gladly honor your request to address you by an alternate name and/or gender pronoun. Please advise me of this early in the semester so that I may make appropriate changes to my records.

### **College-Sponsored Activities and Attendance**

While Colby College is supportive of college-sponsored extracurricular activities such as athletic participation by its students, academics take priority over these activities. For example, both NCAA and Colby rules prohibit missing class for practices. In the case of overlapping commitments between class and college-sponsored extracurricular activities, the student must meet with the professor as soon as possible to discuss these overlaps. The student may request permission to miss class and make up the missed work; the instructor has final authority either to grant or to withhold permission.

### **Religious Holidays**

I have attempted to avoid scheduling exams during major religious holidays. If, however, I have inadvertently scheduled an exam or major deadline that creates a conflict with your religious observances, please let me know within two weeks of the start of classes so that we can make other arrangements. Colby College is supportive of the religious practices of its students, faculty, and staff. The College is committed to ensuring that all students are able to observe their religious beliefs without academic penalty.

### **Sexual Misconduct/Title IX Statement**

Colby College prohibits and will not tolerate sexual misconduct or gender-based discrimination of any kind. Colby is legally obligated to investigate sexual misconduct (including, but not limited to, sexual assault and sexual harassment) and other specific forms of behavior that violate federal and state laws (Title IX and Title VII, and the Maine Human Rights Act). Such behavior also requires the College to fulfill certain obligations under two other federal laws, the Violence Against Women Act (VAWA) and the Jeanne Clery Disclosure of Campus Security Policy and Campus Statistics

Act (Clery Act). To learn more about what constitutes sexual misconduct or to report an incident, review the Student Handbook. I am committed to all Colby students feeling safe, accepted, and included in all aspects of their college experiences, including this course. Colby prohibits and will not tolerate sexual misconduct or gender based discrimination of any kind and is obligated, by federal and state laws, to respond to reports and provide resources to students. As your professor I am considered a “responsible employee” which requires me to report incidence of sexual assault, sexual harassment, dating violence, or stalking to the Title IX Coordinator.

If you wish to access confidential support services, you may contact:

- The Counseling Center: 207-859-4490
- The Title IX Confidential Advocate: 207-509-9122
- The Office of Religious and Spiritual Life: 207-859-4272
- Maine’s 24/7 Sexual Assault Helpline: 1-800-871-7741

### **In-Class Recordings by Students and Unauthorized Distribution of Notes**

Students may only use the notes they take from class for their own personal use or to share with Access and Disabilities Services. Students cannot share or sell these notes via an outside vendor or entity without the instructor’s permission. This pertains to in-class recordings as well. Usage of the notes or in-class recordings in this way without instructor permission is a violation of instructor copyright protection.

### **Inclement Weather**

This course follows the College’s official decisions regarding class cancellations due to inclement weather. If the College cancels classes, our class will not meet. However, I reserve the right to hold class online via Zoom in cases of extenuating circumstances, such as when weather conditions make travel to campus for myself difficult but do not warrant a full College closure. In such cases, I will notify you via email as early as possible with a Zoom link and any adjusted plans for the session.

## **Course Content**

### **Module 1: Introduction to Causality and Regression**

- Unit 1: Introduction
- Unit 2: Good Data Practices and Replication
- Unit 3: Causality and Randomized Controlled Trials
- Unit 4: Ordinary Least Squares (OLS)
- Unit 5: Multivariate Regression
- Unit 6: Statistical Inference

## Module 2: Advanced Modeling Tools

- Unit 7: Dummy Variables and Non-Linearities
- Unit 8: Interaction Terms
- Unit 9: Binary Outcomes

## Module 3: Panel Data and Difference-in-Differences

- Unit 10: Panel Data and Fixed Effects
- Unit 11: Introduction to Difference-in-Differences
- Unit 12: Topics in Difference-in-Differences and Research Design

## Module 4: Other Econometric Methods

- Unit 13: Instrumental Variables
- Unit 14: Introduction to Time Series Methods

## Course Schedule

This schedule is tentative and subject to change.

| Week | Dates        | Topic            | Assignments Due              |
|------|--------------|------------------|------------------------------|
| 1    | Feb 5        | Unit 1           | Concept Check (CC) 1, Survey |
| 2    | Feb 9-13     | Unit 2, Unit 3   | CC 2                         |
| 3    | Feb 16-20    | Unit 3, Unit 4   | CC 3, Problem Set (PS) 1     |
| 4    | Feb 23-27    | Unit 4, Unit 5   | CC 4                         |
| 5    | Mar 2-6      | Unit 5, Unit 6   | CC 5, Topic Proposal         |
| 6    | Mar 9-13     | Unit 6, Midterm  | CC 6                         |
| 7    | Mar 16-20    | Unit 7           | CC 7, PS 2                   |
| 8    | Mar 23-27    | Spring Break     | -                            |
| 9    | Mar 30-Apr 3 | Unit 8, Unit 9   | CC 8, Data Summary           |
| 10   | Apr 6-10     | Unit 10          | CC 9                         |
| 11   | Apr 13-17    | Unit 11          | CC 10, PS 3                  |
| 12   | Apr 20-24    | Unit 12          | CC 11, Preliminary Results   |
| 13   | Apr 27-May 1 | Unit 13, Unit 14 | CC 12                        |
| 14   | May 4-8      | Presentations    | PS 4                         |
| 15   | May 11-15    | Final Exam Week  | Final Draft, Peer Review     |